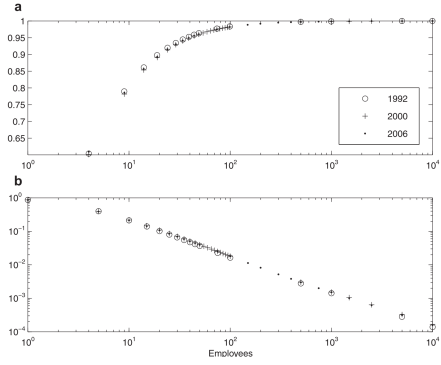
EC991 Topics in Macroeconomics

Week 5 · Firm Dynamics: the Hopenhayn Model

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Many small firms, a few very large employers



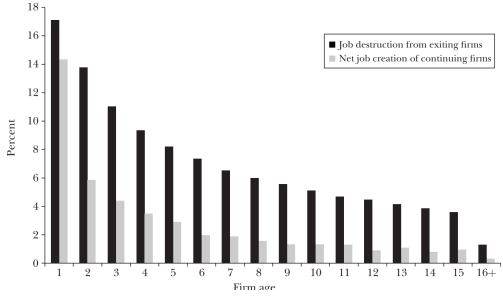
(a) share of US firms with fewer than n employees; (b) share with more than n, log scales; 1992, 2000, 2006

- **Most firms are small**: about 80% of US firms have fewer than 10 employees
- **A few firms are very large**, and they employ a large share of workers
  - half of all employment is in about 18,000 firms
  - a quarter is in the 1,000 or so firms with more than 10,000 employees
- To understand aggregate production, we need to understand **how employment is distributed across firms**
- **The question.** What determines a firm's size, and which firms employ the economy's workers?

Luttmer (2010, Annual Review of Economics), Figure 2, SBA data; Martinez, motivation slides 4 and 7

Firms grow, shrink and exit

Figure 1
Up or Out Dynamics for Young Firms



- **Young firms exit often**: job destruction from exiting firms is about 17% of employment at age 1, and falls with age
- **Young firms that survive grow fast**: net job creation of continuing firms is about 14% at age 1
- The distribution of firms reflects both: changes **within** surviving firms, and **entry and exit**
- These are **rates**: they do not say what share of all jobs young firms create
- **The question.** What determines whether a firm expands, survives or exits?

Decker, Haltiwanger, Jarmin and Miranda (2015, JEP), Figure 1, US data; Martinez, motivation slide 17

Why does the distribution of firms matter?

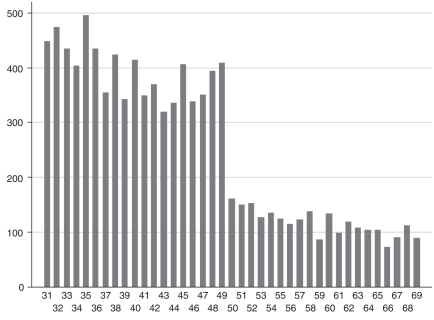


FIGURE 2. NUMBER OF FIRMS BY EMPLOYMENT SIZE IN FRANCE

- Aggregate output depends on firms' productivity **and** on how resources are allocated across them
- Policies can treat firms differently by size. In France, labour regulations become costlier at 50 employees, and the number of firms drops sharply there
- The figure shows firms bunching below the threshold; on its own it does not measure the aggregate cost
- For that we need a model of firm size, entry, exit and aggregate productivity
- **The question.** How do firm-level incentives shape aggregate outcomes?

Garicano, Lelarge and Van Reenen (2016, AER), Figure 2: number of firms by employment, France; Martinez, motivation slides 25 and 27

From households to firms

- Weeks 2 to 4 opened up the **household** side of the aggregate economy. This week we open up the **production** side

	Aiyagari	**Hopenhayn**
Who is heterogeneous	households	firms
Idiosyncratic state	assets and income, (a, y)	productivity, z
Decision	how much to save, a' (continuous)	whether to stay or exit (discrete)
Distribution	$\lambda(a, y)$, sums to one	$\mu(z)$, sums to the **number of firms**
Price found by root finding	interest rate r	output price p
Condition that pins it down	asset market clears	free entry

- The same toolkit: Tauchen, value function iteration, a distribution on a grid, bisection
- One new object: **entry and exit**, which make the number of producers an equilibrium outcome

Introduction

- **Goal.** Present the canonical model of **industry dynamics**: Hopenhayn (1992)
- It underpins a wide range of applications:
  - business cycles: investment, employment, adjustment costs, financial shocks
  - development and growth: misallocation, financial development
  - international trade, labour, and more
- We first build intuition in a **static** model, then move to the quantitative, dynamic one

A simple Hopenhayn economy

- A fixed measure M of firms, indexed by i: no entry or exit yet
- A fixed number of workers, N
- Firms (plants) differ in productivity, $z \sim G(z)$
- Labour is the only input, with **decreasing returns**:

$$y = z n^{\eta}, \qquad 0 < \eta < 1$$

- The wage w adjusts until firms' total labour demand equals the fixed supply N

A simple Hopenhayn economy: the firm

- With the price of output normalised to one, each firm maximises profit:

$$\pi(z) \;=\; \max_{n} \; \left\{ z\, n^{\eta} - w\, n \right\}$$

- The first-order condition for firm i: $\eta\, z_i\, n_i^{\eta-1} = w$
- Every firm faces the same wage, so the **marginal product of labour** is the same everywhere. For any two firms i and j:

$$\eta\, z_j\, n_j^{\eta-1} \;=\; \eta\, z_i\, n_i^{\eta-1} \qquad \Longleftrightarrow \qquad \frac{z_j}{z_i} \;=\; \left(\frac{n_j}{n_i}\right)^{1-\eta}$$

- a firm twice as productive hires $2^{1/(1-\eta)}$ times as many workers: 8 times when $\eta = 2/3$

The efficient allocation

- The market allocation is also **efficient**. A planner who maximises output chooses

$$\max_{\{n_i\}} Y = \int y_i \, di = \int z_i \, n_i^{\eta} \, di \qquad \text{s.t.} \qquad N = \int n_i \, di$$

- With multiplier μ on the labour constraint, the first-order condition for firm i is

$$\eta \, z_i \, n_i^{\eta - 1} = \mu \qquad \Longrightarrow \qquad \frac{z_j}{z_i} = \left(\frac{n_j}{n_i} \right)^{1-\eta}$$

- **Efficiency equalises the marginal product of labour across producers**
  - more productive firms (higher z) should employ more workers

Aggregation

- With $y = zn^{\eta}$, equal marginal products also mean equal **average products**:

$$\frac{y_i}{n_i} = z_i n_i^{\eta-1} = \frac{\mu}{\eta}$$

- So aggregate output is proportional to aggregate labour:

$$Y = \int z_i n_i^{\eta} \, di = \frac{\mu}{\eta} \int n_i \, di = \frac{\mu}{\eta} N$$

- Solving the first-order condition for n_i and using the resource constraint:

$$N = \int n_i \, di = \int \left(\frac{\eta z_i}{\mu} \right)^{\frac{1}{1-\eta}} di \qquad \Longleftrightarrow \qquad \frac{\mu}{\eta} N = \left(\int z_i^{\frac{1}{1-\eta}} \, di \right)^{1-\eta} N^{\eta}$$

The aggregate production function

- The aggregate production function has the **same form** as each firm's technology:

$$Y = Z N^{\eta}, \qquad Z = \left(\int z_i^{\frac{1}{1-\eta}} \, di \right)^{1-\eta}$$

- It helps to write it in terms of the productivity distribution and the number of firms M:

$$Y = \left(\int z^{\frac{1}{1-\eta}} \, dG(z) \right)^{1-\eta} M^{1-\eta} N^{\eta}$$

- Read this way, aggregate technology has **constant returns in** M **and** N
- TFP is a **power mean** of firm productivity, of order $1/(1-\eta)$: it weights productive firms more than their number, because they hire more

The efficient allocation: an example

- Two firms, $z_1 = 1$ and $z_2 = 2$, with $\eta = 2/3$ and $N = 9$ workers
- **Efficient split.** $n_2/n_1 = (z_2/z_1)^{1/(1-\eta)} = 2^3 = 8$, so $n_1 = 1$ and $n_2 = 8$:

$$Y \;=\; 1 \times 1^{2/3} + 2 \times 8^{2/3} \;=\; 1 + 8 \;=\; 9$$

- **Aggregate TFP.** $Z = \left(1^3 + 2^3\right)^{1/3} = 9^{1/3} \approx 2.08$, and $ZN^{\eta} = 9^{1/3} \times 9^{2/3} = 9$: the same number
- **An equal split instead.** $n_1 = n_2 = 4.5$:

$$Y \;=\; (1+2) \times 4.5^{2/3} \;\approx\; 8.18$$

- The same workers and the same firms produce 9.1% less: the marginal product is higher at firm 2, so moving workers there raises output
- This loss is **misallocation**. In the notation of the next slides, $\Omega = 8.18/9 \approx 0.909$

Aggregation: a benchmark

- The aggregation result is a useful **benchmark**
- Aggregate output moves with the number of firms and with the distribution of productivity
- It extends to several inputs. With $y_i = z_i f(k_i, n_i)^\eta = z_i \left(k_i^\alpha n_i^{1-\alpha}\right)^\eta$:

$$Y = \left(\int z_i^{\frac{1}{1-\eta}} di \right)^{1-\eta} f(K, N)^\eta$$

- where K is aggregate capital
- Efficiency now also requires the **marginal product of capital** to be equal across producers

Misallocation wedges

- Now let each firm face a **wedge** τ_i: a subsidy if $\tau_i > 0$, a tax if $\tau_i < 0$

$$\pi(z_i) \;=\; \max_{n_i} \left\{(1+\tau_i)\, z_i\, n_i^{\eta} - w\, n_i\right\}$$

- The first-order condition becomes $\eta\,(1+\tau_i)\, z_i\, n_i^{\eta-1} = w$
- Marginal products are **no longer equal** across firms:

$$(1+\tau_j)\,\eta\, z_j\, n_j^{\eta-1} \;=\; (1+\tau_i)\,\eta\, z_i\, n_i^{\eta-1} \qquad \Longleftrightarrow \qquad \frac{(1+\tau_j)\, z_j}{(1+\tau_i)\, z_i} \;=\; \left(\frac{n_j}{n_i}\right)^{1-\eta}$$

- a subsidised firm hires as if it were more productive than it is

Aggregation with wedges

- Aggregate output and labour, using each firm's labour demand:

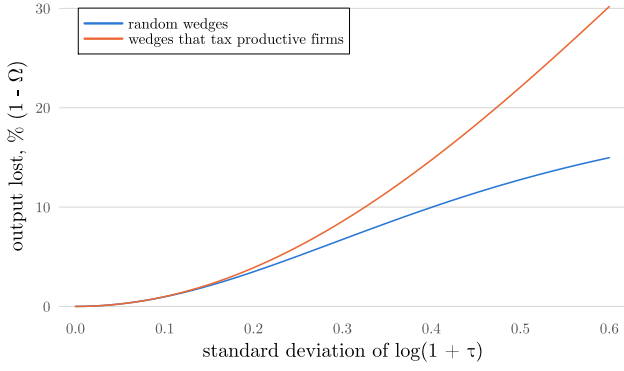
$$Y = \left(\frac{\eta}{w}\right)^{\frac{\eta}{1-\eta}} \int z_i^{\frac{1}{1-\eta}} (1+\tau_i)^{\frac{\eta}{1-\eta}} \, di, \qquad N^\eta = \left(\frac{\eta}{w}\right)^{\frac{\eta}{1-\eta}} \left(\int z_i^{\frac{1}{1-\eta}} (1+\tau_i)^{\frac{1}{1-\eta}} \, di\right)^\eta$$

- Dividing one by the other: $Y = Z \, \Omega \, N^\eta$, where
  - $Z = \left(\int z_i^{1/(1-\eta)} \, di\right)^{1-\eta}$ is **efficient TFP**, as before
  - $\Omega \in (0, 1]$ is the **misallocation term**:

$$\Omega = \frac{\int z_i^{\frac{1}{1-\eta}} (1+\tau_i)^{\frac{\eta}{1-\eta}} \, di}{\left(\int z_i^{\frac{1}{1-\eta}} (1+\tau_i)^{\frac{1}{1-\eta}} \, di\right)^\eta \left(\int z_i^{\frac{1}{1-\eta}} \, di\right)^{1-\eta}}$$

- If every firm faces the same wedge, $\Omega = 1$: a common wedge only moves the wage. What costs output is **dispersion** in wedges

How much output do wedges cost?



- $\eta = 2/3$, productivity from the stationary distribution of the dynamic model below
- **Random wedges**: at every z, half the firms get $\log(1+\tau) = +s$ and half $-s$
- **Wedges that tax productive firms**: $\log(1+\tau)$ falls with $\log z$, with the same dispersion. They do more damage, because they shrink exactly the firms that should be large

Monopolistic competition

- An alternative, following Melitz (2003): **monopolistic competition**
- A final good aggregates a continuum of intermediate varieties:

$$Y = \left(\int_0^M y_i^{\eta} \, di \right)^{\frac{1}{\eta}}, \qquad 0 < \eta < 1 \quad \text{(gross substitutes)}$$

- The standard demand for each variety and the price index:

$$y_i = \left(\frac{p_i}{P} \right)^{\frac{1}{\eta - 1}} Y, \qquad P = \left(\int p_i^{\frac{\eta}{\eta - 1}} \, di \right)^{\frac{\eta - 1}{\eta}}$$

Monopolistic competition: the producers

- Each intermediate producer has a linear technology, $y_i = \tilde{z}_i \, n_i$, with $\tilde{z}_i = z_i^{1/\eta}$
- As a monopolist, it chooses its price and quantity together:

$$\max_{y_i,\, p_i} \; p_i \, y_i - w \, \frac{y_i}{\tilde{z}_i} \qquad \text{s.t.} \qquad y_i \;=\; \left(\frac{p_i}{P}\right)^{\frac{1}{\eta-1}} Y$$

- The price is a constant **markup** over marginal cost:

$$p_i \;=\; \frac{1}{\eta} \, \frac{w}{\tilde{z}_i}$$

- More productive firms charge less and win a larger share of the market
  - so they earn more revenue and more profit

Monopolistic competition: aggregation

- Some algebra (see Melitz) gives

$$Y = \left(\int \tilde{z}^{\frac{\eta}{1-\eta}} \, dG(z) \right)^{\frac{1-\eta}{\eta}} M^{\frac{1-\eta}{\eta}} N \qquad \Longleftrightarrow \qquad Y^{\eta} = \left(\int z^{\frac{1}{1-\eta}} \, dG(z) \right)^{1-\eta} M^{1-\eta} N^{\eta}$$

- The aggregate production function in Melitz is a **monotone transformation** of Hopenhayn's: whatever maximises Y also maximises Y^{η}
- The difference: in Melitz, efficiency equalises the **marginal revenue product** of labour across firms
  - in Hopenhayn every firm sells at the same price; in Melitz prices differ across firms
  - this matters when we take the model to data: we usually observe revenue, not quantities

Entry

- Suppose opening a new firm takes c_e workers
  - once created, the firm draws its z from $G(z)$: heterogeneity is revealed **ex post**
  - an alternative is **ex-ante** heterogeneity: the firm sees z first and then decides whether to enter
- How would a planner choose the **number of firms**?
- In two steps:
  1. for a given number of firms, split the production workers efficiently across them (what we just did)
  2. choose the number of firms

Entry: creating firms or producing goods?

- Allocate N workers between creating firms (N_e) and producing goods (N_p). Since each firm requires c_e workers to create:

$$M = \frac{N_e}{c_e}, \qquad Y = Z\left(\frac{N_e}{c_e}\right)^{1-\eta} N_p^{\eta}, \qquad N_e + N_p = N$$

- **What happens if we move a little labour from production to entry?** The extra firms raise output, but fewer production workers lower it:

$$\underbrace{\frac{\partial Y}{\partial N_e} = (1-\eta)\frac{Y}{N_e}}_{\text{marginal gain from entry}}, \qquad \underbrace{\frac{\partial Y}{\partial N_p} = \eta \frac{Y}{N_p}}_{\text{production forgone}}$$

- At the optimum, this reallocation cannot improve output:

$$\frac{\partial Y}{\partial N_e} - \frac{\partial Y}{\partial N_p} = 0 \qquad \Longrightarrow \qquad \frac{1-\eta}{N_e} = \frac{\eta}{N_p}$$

- Together with $N_e + N_p = N$, this gives:

$$\boxed{N_e = (1-\eta)N, \qquad N_p = \eta N, \qquad M = \frac{(1-\eta)N}{c_e}}$$

- **Decreasing returns ($\eta < 1$) explain the benefit of additional firms**: spreading workers across more producers mitigates diminishing returns. Stronger decreasing returns favour more entry; weaker decreasing returns favour expanding existing firms

Entry: what it implies

- Substituting the optimal number of firms and production employment:

$$Y = Z \eta^{\eta} (1-\eta)^{1-\eta} c_e^{-(1-\eta)} N$$

- The elasticity of output per worker with respect to the entry cost is $-(1-\eta)$

- **Aggregate productivity reflects both firm technologies and the cost of creating firms**
  - higher entry costs $\rightarrow$ fewer firms $\rightarrow$ more workers per firm
  - decreasing returns then imply lower output per worker

- **The same firm productivity distribution can generate different output per worker**
  - differences in the cost of establishing businesses are one possible explanation

- With $\eta = 2/3$, doubling c_e lowers output per worker by approximately **21%**, even though firms' technologies are unchanged

The dynamic model

The dynamic model: Hopenhayn (1992)

- So far everything is **static**: productivity never changes, so firms never grow or shrink
- Make productivity stochastic, and we have the **workhorse model of industry dynamics**
  - we study the **stationary equilibrium**: firms enter, grow, shrink and exit, while the distribution of firms stays the same
  - the **firm size distribution** is an equilibrium outcome
- The household side is kept very simple for now: a demand curve

The environment

- A continuum of firms, each of measure zero, with decreasing returns: $y_i = z_i \, n_i^{\eta}$
- **Idiosyncratic risk**: each firm's productivity z follows a first-order Markov process, $F(z' \mid z)$
- **Entrants** draw their first productivity from a fixed distribution, $z_0 \sim G(z)$
  - letting G differ from incumbents' distribution allows a richer firm size distribution
- Two costs, both paid in labour: an **entry cost** c_e and a **fixed cost** c_f every period a firm operates
- At the start of every period, incumbents decide whether to **stay or exit**, and potential entrants whether to **enter**

Incumbent firms: the static problem

- Each period, an incumbent maximises profit at output price p and wage w:

$$\pi(z;\, p, w) \;=\; \max_{n} \;\left\{ p\, z\, n^{\eta} - w\, n - w\, c_f \right\}$$

- The familiar solution:

$$\eta\, p\, z\, n^{\eta-1} \;=\; w \qquad \Longrightarrow \qquad n(z;\, p, w) \;=\; \left(\frac{\eta\, p\, z}{w}\right)^{\frac{1}{1-\eta}}$$

- and profit:

$$\pi(z;\, p, w) \;=\; (1-\eta)\,(p\, z)^{\frac{1}{1-\eta}} \left(\frac{\eta}{w}\right)^{\frac{\eta}{1-\eta}} - w\, c_f$$

- With $c_f > 0$, there is a z at which $\pi = 0$. From now on, **the wage is the numeraire**, $w = 1$, and we solve for the equilibrium price p

Incumbent firms: the exit decision

- Today's operation is already committed: the firm observes z and earns $\pi(z;\, p)$
- It then decides whether to operate **next period**, before observing z'
- With discount factor $\beta = 1/(1+r)$, its value is:

$$V(z) \;=\; \underbrace{\pi(z;\, p)}_{\text{today's profit}} \;+\; \beta \,\max\Big\{ \underbrace{\int V(z')\, dF(z' \mid z)}_{\text{operate next period}},\; \underbrace{0}_{\text{exit: no future payoff}} \Big\}$$

- Today's profit is **outside the maximum**: exiting does not avoid today's operating loss
- $V(z')$ includes next period's profit **and the option to exit thereafter**
- There is no **scrap value**: no net proceeds from selling assets when the firm closes
- The policy is **discrete**: $\chi(z) = 1$ means exit; $\chi(z) = 0$ means continue

Incumbent firms: the exit threshold

- Profit rises with z, and with a persistent (monotone) F, so does the value function
- So there is a **threshold** $\tilde{z}$: every firm with $z < \tilde{z}$ exits
- At the threshold, the expected value of staying equals the value of exit:

$$\mathbb{E}\left[V(z') \,\middle|\, \tilde{z}\right] \;=\; \int V(z')\, dF(z' \mid \tilde{z}) \;=\; 0$$

- This does **not** mean firms never make losses: a firm may stay with $\pi(z) < 0$ today if it expects z to recover

Entrants and free entry

- Potential entrants are identical **ex ante**
- An entrant pays $c_e > 0$ to set up a plant and draws $z \sim G(z)$; it starts producing next period
- The value of entering:

$$V^e \;=\; -c_e \;+\; \beta \int V(z)\, dG(z)$$

- Firms enter as long as $V^e \geq 0$. If $V^e > 0$, entry continues until the value is driven to zero: **free entry**
- Let $M \geq 0$ be the mass of entrants. In equilibrium, free entry requires

$$\beta \int V(z)\, dG(z) \;\leq\; c_e, \qquad \text{with equality if } M > 0$$

- For some parameters there may be **no entry**, $M = 0$: then entering must not pay, $V^e < 0$

The distribution of firms

- Let $\mu_t([0, z])$ be the **measure** of firms with productivity up to z
- Entry and exit decisions give its law of motion:

$$\mu_{t+1}(z') = \underbrace{\int F(z' \mid z)\,(1 - \chi(z))\,d\mu_t(z)}_{\text{incumbents who stay, after their shock}} + \underbrace{M_{t+1}\,G(z')}_{\text{entrants}}$$

- In a **stationary equilibrium**, $\mu_{t+1} = \mu_t = \mu$
- As in Aiyagari, the distribution is constant while individuals move: firms keep changing size as z moves, and keep entering and exiting
- Unlike Aiyagari, μ does not sum to one: its total is the **number of firms**, an equilibrium object

Demand and supply

- Households demand the good. To keep things simple, take demand as given by a function $D(p)$, with $D'(p) < 0$
  - a simple form: $D(p) = \bar{D}/p$, so spending $pD(p) = \bar{D}$ is fixed
- Supply comes from the firms that operate:

$$Y(p) \;=\; \int y(z;\, p)\, d\mu(z)$$

  the costs c_e and c_f are paid in labour, so they do not appear here
- The goods market clears: $D(p) = Y(p)$
  - $Y(p)$ rises with the price; $D(p)$ falls with it

Stationary equilibrium

- A **stationary recursive competitive equilibrium** is a price p, a mass of entrants M, an exit threshold $\tilde{z}$ and a measure μ such that:
  - the goods market clears
  - incumbents' exit decisions are optimal
  - there is no further incentive to enter
  - μ reproduces itself under the law of motion
- The main difference from Aiyagari: the **number of firms** is also endogenous
- But the law of motion is **linear** in μ and M, so we can find p without knowing M, and solve in two blocks:
  1. the price p, from free entry alone
  2. the mass of entrants M, from the goods market

Computing the equilibrium

Solving for the equilibrium: the price

- Discretise z on n_z grid points; the usual methods apply (Tauchen, week 3)
  - there is a **discrete choice**, so do not skimp on grid points: we will see why in the tutorial
- Suppose $M > 0$. Find the price p in four steps:
  - **Step 1.** Guess a price p^0. Compute $\pi(z_i;\, p^0)$, $n(z_i;\, p^0)$ and $y(z_i;\, p^0)$ at every grid point
  - **Step 2.** Solve the incumbent's Bellman equation by value function iteration
  - **Step 3.** With the value function $V(z)$, check the free entry condition
  - **Step 4.** If free entry fails, update the guess and return to Step 1

Step 2: value function iteration

- Let i index the grid for z, and F_{ij} be the probability of moving from z_i to z_j
- Guess $V^0(z_i)$, an $n_z \times 1$ vector. Update it at every grid point:

$$V^1(z_i;\, p^0) \;=\; \pi(z_i;\, p^0) \;+\; \beta \,\max\Big\{ \sum_{j=1}^{n_z} F_{ij}\, V^0(z_j;\, p^0),\; 0 \Big\}, \qquad i = 1, \ldots, n_z$$

- If $\max_i |V^1(z_i) - V^0(z_i)| < \text{tol}$, stop. Otherwise set $V^0 = V^1$ and repeat
- In matrices, the whole update is **one product** FV^0 and an element-wise maximum
- Once V has converged, record the exit decision as an $n_z \times 1$ vector:

$$\chi(z_i;\, p^0) \;=\; \begin{cases} 1 & \text{if } \sum_j F_{ij}\, V(z_j;\, p^0) < 0 \\ 0 & \text{otherwise} \end{cases}$$

Steps 3 and 4: free entry and the price

- Let g_i be the probability that an entrant draws z_i. The value of entry:

$$V^e(p^0) \;=\; -c_e \;+\; \beta \sum_{i=1}^{n_z} g_i \, V(z_i;\, p^0)$$

- With $M > 0$, free entry requires $V^e(p^0) = 0$
- $V(z_i;\, p)$ **rises with** p, so $V^e(p)$ does too: a root-finding problem in one variable, solved by bisection or Brent, as for r in week 4
  - if $V^e(p^0) > 0$: entry is too attractive. **Lower** the price
  - if $V^e(p^0) < 0$: entry does not pay. **Raise** the price
- With the new guess p^1, recompute $\pi(z;\, p^1)$ and $V(z;\, p^1)$, until $V^e = 0$

Step 5: the distribution of firms

- With the price found, the law of motion of μ and the goods market give M
- Let μ_i be the mass of operating firms at productivity z_i. Their total mass is

$$B \;=\; \sum_i \mu_i$$

- B **is the stock of operating firms;** M **is the flow of entrants per period**
- Write $\hat{F}$ for F with the rows of exiting firms set to zero:

$$\hat{F}_{ij} \;=\; \left(1 - \chi(z_i;\, p)\right) F_{ij}$$

- The stationary law of motion is **linear**, and solves in one step:

$$\mu \;=\; \hat{F}'\mu + Mg \qquad \Longrightarrow \qquad \mu \;=\; M\,(I - \hat{F}')^{-1} g$$

- μ and g are $n_z \times 1$ vectors, $\hat{F}$ is $n_z \times n_z$. $(I - \hat{F}')$ is invertible because some firms exit every period
- The stationary distribution depends on the price and the mass of entrants: $\mu(p, M)$, and it is **proportional to** M
- In Aiyagari we iterated $\lambda_{t+1} = Q'\lambda_t$ to convergence; iterating here also works, and gives the same μ

Step 6: the mass of entrants

- To find M, clear the goods market. Supply is the output of every operating firm:

$$D(p) \;=\; \sum_{i=1}^{n_z} y(z_i;\, p)\; \mu(z_i;\, p, M)$$

- p is known, so this is one equation in M
- **The trick**: μ is linear in M, so $\mu(p, M) = M \times \mu(p, 1)$. Hence

$$M \;=\; \frac{D(p)}{\sum_{i=1}^{n_z} y(z_i;\, p)\; \mu(z_i;\, p, 1)}$$

- Recover the equilibrium firm measure and total firm mass:

$$\mu_i \;=\; M\bar{\mu}_i, \qquad B \;=\; M \sum_i \bar{\mu}_i$$

  where $\bar{\mu} = \mu(p, 1)$ is the stationary measure computed with $M = 1$

- If $M > 0$, we have found the equilibrium $(p, M, \tilde{z}, \mu)$
- No loop over M at all: once p is known, everything else follows in closed form

An alternative: a stationary industry with no entry

- The previous algorithm seeks an equilibrium with **positive entry and exit**
- Could there instead be a stationary equilibrium with **no entry**, $M = 0$?
- With no entrants replacing departing firms, stationarity also requires **no exit**
- **Construct a candidate equilibrium**
  - Take the existing firm mass $B > 0$ as given
  - Compute the invariant productivity probabilities f:

$$f = F'f, \qquad \sum_i f_i = 1, \qquad \mu_i = Bf_i$$

  - Find the price that clears the goods market:

$$D(p) = B \sum_i y(z_i; p) f_i$$

- **Verify that the candidate is an equilibrium**
  - Solve the firm's value function at this price
  - Check that **incumbents choose to stay** at every occupied productivity state
  - Check that **entry is unattractive or just breaks even**: $V^e(p) \leq 0$
  - If either check fails, this is **not** a stationary no-entry equilibrium for the chosen B

What the model delivers

Example: parameters

- Log productivity follows an AR(1) around a mean of 1:

$$\log z' - 1 \;=\; \rho\,(\log z - 1) + \sigma\,\varepsilon', \qquad \varepsilon' \sim N(0,1)$$

- Tauchen with $n_z = 101$ points, ± 4 standard deviations
- Entrants draw from the **stationary distribution** of the incumbents' chain, $g = f$

Parameter	Value	Role
η	$2/3$	curvature of the production function
β	0.8	discount factor
c_e	40	entry cost, in workers
c_f	20	fixed cost per period, in workers
$\bar{D}$	100	demand, $D(p) = \bar{D}/p$
ρ, σ	0.9, 0.2	persistence and volatility of productivity

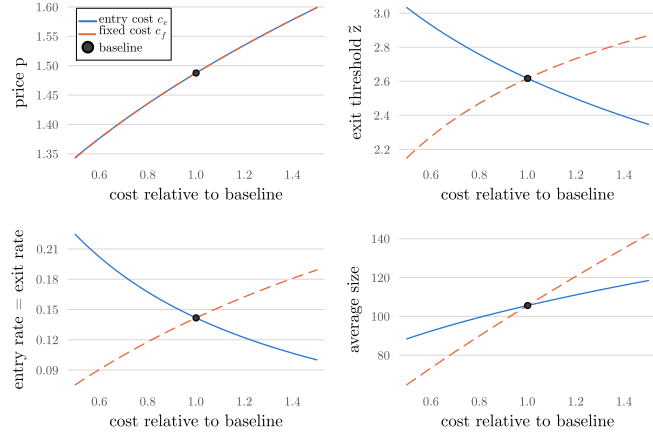
Comparative statics: a higher entry cost protects incumbents

- **Entry must still break even.** A higher setup cost requires higher expected operating profits, so the equilibrium output price rises
- **Incumbents benefit without paying the higher setup cost again.** Higher revenues let weaker firms survive: the exit threshold falls
- **Entry becomes harder, but survival becomes easier.** Entrants still draw productivity from the same G; selection through exit becomes less strict
- **Longer-lived firms need replacing less often.** Entry and exit rates fall, and the operating population becomes older
- **Average firm size faces two opposing forces:** higher prices expand firms at each productivity, while weaker selection retains more small firms. Here, expansion dominates
- **Fewer firms meet lower demand:** the higher price reduces aggregate demand, while average output per firm rises. Thus B falls even though firms survive longer

Numerical example: raise c_e from 40 to 60

Outcome	Baseline	Higher entry cost
Price	1.486	1.597
Exit threshold	2.619	2.348
Entry rate = exit rate	13.4%	10.6%
Mean firm age	11.5 years	13.7 years
Operating firm mass B	0.641	0.553
Average employment	104.0	120.6
Aggregate output	67.29	62.60

Comparative statics: entry cost against fixed cost



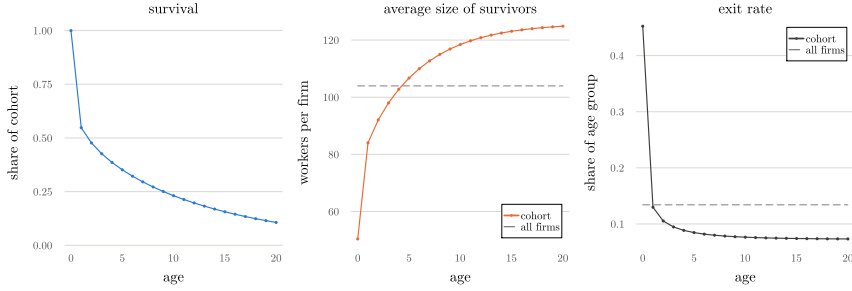
- Both costs raise the price, by almost the same amount here, but they move **selection** in opposite directions. The price alone cannot tell us which cost is high; turnover can

$n_z = 1001$ **with the exit lottery; each cost scaled from 0.5 to 1.5 times its baseline**

Other empirical issues

- Employment is tied one for one to productivity, so size and productivity move together: a small productivity shock moves workers between firms
- **Unconditionally**, a firm's age matters:
  - firms enter small (given the assumption on the entrants' distribution), and survive only if they draw high productivity, which makes them larger
  - so the model predicts that large firms are old and more productive
  - but **conditional on size**, age does not matter: z is the only state
- Only small firms exit. In the data, some large firms exit too

A cohort of entrants



- 45% of entrants exit after their first year; a quarter of the cohort is still producing at age 9
- Entrants average 50 workers, half the industry average. Survivors grow to 125 by age 20. No firm grows on purpose: this is **selection**, as low draws leave
- Exit rates fall with age, as in the data. Here entrants draw from the same distribution as incumbents; with a G that puts entrants lower, they start smaller still

Conclusion

- The firm dynamics model opens the **black box** of the aggregate production function
- The model here is **efficient**: the welfare theorems hold, and the competitive equilibrium solves the planner's problem
  - policies, such as taxes, break this and can distort the allocation of employment
- We have left out capital. Adding capital **without frictions** changes nothing essential
- Many papers add capital **with frictions**. Early examples:
  - Veracierto (2002, AER): irreversible plant-level investment and the aggregate propagation of individual investment
  - Cooley and Quadrini (2001, AER) and Gomes (2001, AER): firms facing financial frictions

Where to go from here

- **Capital frictions**: Veracierto (2002, AER); Cooley and Quadrini (2001, AER); Gomes (2001, AER); Cooper and Haltiwanger (2006, REStud)
- **Labour market frictions**: Hopenhayn and Rogerson (1993, JPE); Fujita and Nakajima (2016, RED); Kaas and Kircher (2015, AER); Bilal et al. (2022, Econometrica)
- **Innovation**: Klette and Kortum (2004, JPE); Akcigit and Kerr (2018, JPE)
- **Development, firm size and informality**: Poschke (2018, AEJ: Macro); Bento and Restuccia (2017, AEJ: Macro); Ulyssea (2018, AER)
- **International trade and open economies**: Melitz (2003, Econometrica); Coşar et al. (2016, AER); Kambourov (2009, REStud); Edmond et al. (2015, AER); Dix-Carneiro et al. (2021, WP); Salomao and Varela (2022, REStud)
- **Demographics, declining dynamism and growth**: Hopenhayn et al. (2022, Econometrica); Pugsley et al. (forthcoming, AER); Asturias et al. (2023, AEJ: Macro)
- **Entrepreneurial heterogeneity**: Queiró (2022, REStud); Yurdagul (2017, JME)

References

- **Hopenhayn (2014, Annual Review of Economics)***: a broad survey, and an accessible read
- **Hopenhayn (1992, Econometrica)**: the original paper; much of it sets out the mathematical foundations
- **Hopenhayn and Rogerson (1993, JPE)***: the best-known application, to firing taxes
- Chris Edmond's lecture notes on firm dynamics are another good source

* core reading